



EXPLOITS UNIVERSITY

BSc INFORMATION COMMUNICATION AND TECHNOLOGY

BICT 3202 · OBJECT-ORIENTED ANALYSIS AND DESIGN

WEEK 16

Assignment: Comprehensive Revision and Examination Preparation

PROGRAMME	BSc Information Communication and Technology
COURSE CODE / YEAR	BICT 3202 · Year 3
LECTURER	Francis Fweta — ICT Lecturer
DELIVERY	2h Lecture · 1h Tutorial · 1h Practical · 10h Independent Learning

Prepared by Francis Fweta · Exploits University · Week 16 of 16



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ASSIGNMENT INSTRUCTIONS

- Answer ALL questions in Sections A, B and C.
- Read the Week 16 module and your lecture notes before attempting each question.
- This assignment doubles as your examination revision.
- Draw UML diagrams in correct notation, and check that your models are consistent.
- Submit by the date given by your lecturer. Total: 50 marks.

Section A — Short-Answer Questions (20 marks)

Question A1

Define object-oriented analysis and object-oriented design. [3 marks]

Question A2

Distinguish an object from a class. [2 marks]

Question A3

Define encapsulation and abstraction, with one example each. [3 marks]

Question A4

Define inheritance and polymorphism. [2 marks]

Question A5

Explain the business-to-software chain from Week 1. [3 marks]

Question A6

State the four principles of object modelling. [2 marks]

Question A7

What is UML and what is its current specification version? [2 marks]

Question A8

Explain the relationship between the Object, Dynamic and Behavioural models. [3 marks]

Section B — Application Questions (20 marks)

Question B1 (12 marks)

Consider the university course registration system as a comprehensive case study.

- (a) Apply the business-to-software chain to this system. [4 marks]
- (b) Identify all objects, attributes, behaviours and relationships. [4 marks]
- (c) Draw the complete object model. [4 marks]

Question B2 (8 marks)

Consider the integration of all 16 weeks of OOAD concepts.

- (a) Explain how all concepts fit together. [4 marks]
- (b) Identify the most important lessons from the module. [4 marks]

Section C — Case Study (10 marks)

Consider a new system of your choice (e.g., e-commerce, hospital, banking).

As an object-oriented analyst/designer:

- (a) Apply OOAD concepts from Weeks 1-16 to analyse and design this system. [5 marks]
- (b) Explain how the concepts support a complete and effective design. [3 marks]
- (c) Identify lessons learned from the OOAD module. [2 marks]

Marking Scheme

Section	Content	Marks
A	Short-answer questions (8 questions)	20
B	Application questions (12 + 8)	20
C	Case study — comprehensive system	10
	TOTAL	50

Submission & Academic Integrity

- Submit a typed document (or neatly handwritten, as directed by your lecturer).
- This assignment is individual work. Plagiarism or copying will attract a penalty under university regulations.
- Ensure your answers demonstrate comprehensive understanding of OOAD.
- Cite any sources you consult using the format used in the Week 16 module references.